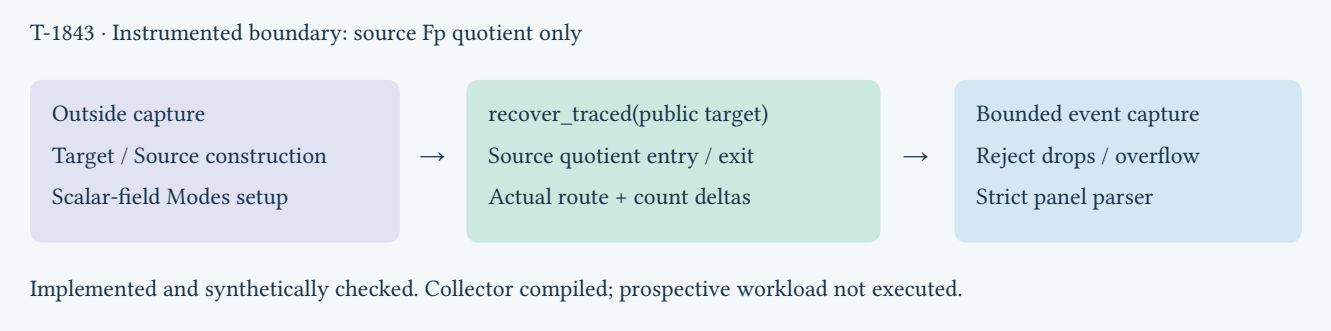
Source quotient trace: a precise measurement boundary

T-1843 · 2026-11-07 · Instrument construction and synthetic validation

Implemented an opt-in trace for the weighted quotient kernel inside the public-point source over Fp. Code inspection separates this from Modes' polynomial kernel over scalar modulus r. Keeping those layers separate is necessary before attributing source costs or selecting further optimizations.



Recorded event. Source modulus, quotient length, divisor degree, full relevant support size and weighted support cost, actual branch, quotient-only field products and inversion entries, and success. Branch markers are inside the implementation. Counter snapshots exclude remainder multiplication. Audit support scans and trace allocation are overhead; no elapsed-time interpretation is made.

Integrity. Same-thread begin/finish, no nested capture, checked counters, at most 100,000 records per target, explicit invalid status on drops or overflow. Normal build excludes trace module and exports. Public collector inputs remain points; scalar indices are only target-generation/reference labels.

Software check	Coverage	Outcome
Native synthetic cases	120	Reference equality; metadata; routes
Exact sparse counts / A/A	Accepted synthetic paths	Pass
Record cap / overflow	Forced failures	Detected; capture invalid
Thread isolation / lifecycle	Same-thread controls	Pass
Parser fabricated panel	617 target headers	Accepted as synthetic test only
Parser malformed panels	6 mutations	All rejected
Collector / normal source	Build only	Pass; collector not executed

Prospective panel. The valid draft protocol covers 617 targets from five existing prime-field fixtures, fresh Source per target, two independent identical processes, and exact transcript equality. Proposed shared ceiling: one attempt, 120 seconds, one CPU, 1024 MiB, USD 1; zero funds reserved. It is not frozen or admitted. Published source/binary binding, canonical execution/export adapter, admitted preflight and current budget/quota evidence are required before data collection. The environment opt-in is not authorization.

Preserved failures. Two receipt-wrapper filename errors followed successful native builds; the synthetic binary name was corrected and the normal metadata artifact reconciled without recompilation. Original wrappers/failures remain. System jsonschema lacked the required validator; the existing virtualenv validates the corrected draft. No workload data or comparative result was produced.

Evidence and next step. README.md defines the boundary and prospective fixtures. Native receipts and generator pins retain provenance. PCC-20261107-7f4842 is unsigned producer evidence. Original 4,563-case cohort unchanged. Next complete the admission/export adapter, then collect the frozen panel when its existing execution requirements are satisfied. Sources: exact predecessor and recovery modules, pinned protocol authority, FLINT series division; KB unavailable.